

L2: ML : Linear Regression - 1

↑
Supervised Learning

Linear Regression



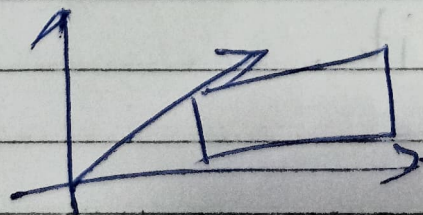
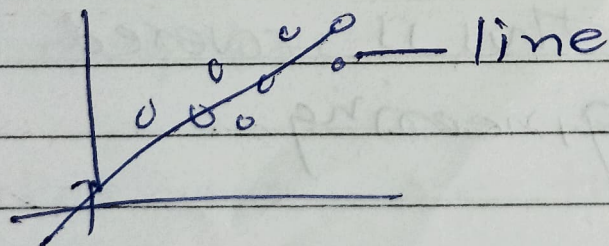
line

$$y = mx + c$$

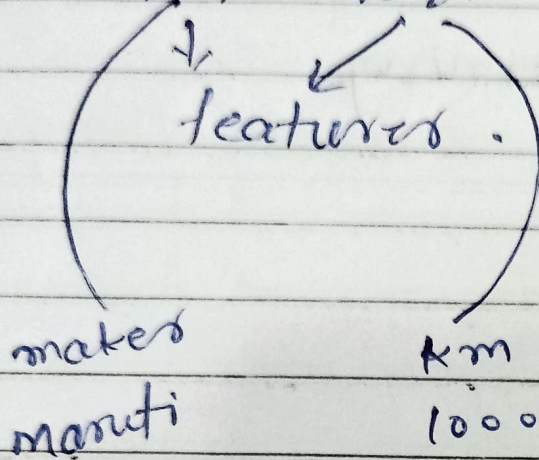
$$y = m_1x_1 + m_2x_2 + \dots + c$$

line passing through points

hyperplane passing through points



$$y = m_1 x_1 + m_2 x_2 + \dots + C$$



$x_1 \rightarrow$ makes

$x_2 \rightarrow$ km

we can do $m_2 x_2$ but not $m_1 x_1$
as x_1 is string

so we need to ~~conv~~ encode x_1
to numbers. This is covered in
feature Engineering.

↓
Need to study it.

Encoding $\rightarrow$ onehot
label
target

Onehot encoding $\Downarrow$

	Moruti	Tata	Bata
Moruti	1	0	0
Tata	0	1	0
Moruti	1	0	0
Bata	0	0	1

problem with this
too many cols

if makes are $4k$ then $4k$ cols
needs

label encoding

T - 0

B - 1

M - 2

Z - 3

S - 4

problem: $\rightarrow$ high numbers such as 4 have high weightage / preference.

where it is used

if I want to encode degree

B.Tech - 0

M.Tech - 1

p.hd - 2

P Phd - 3

it is ok to use label encoding as it is ordinal data

Target encoding

make	Target price
M	30
S	20
M	12
B M	10
S	11
T	13

take avg. at M, S, & T.

$$M \rightarrow 17.5$$

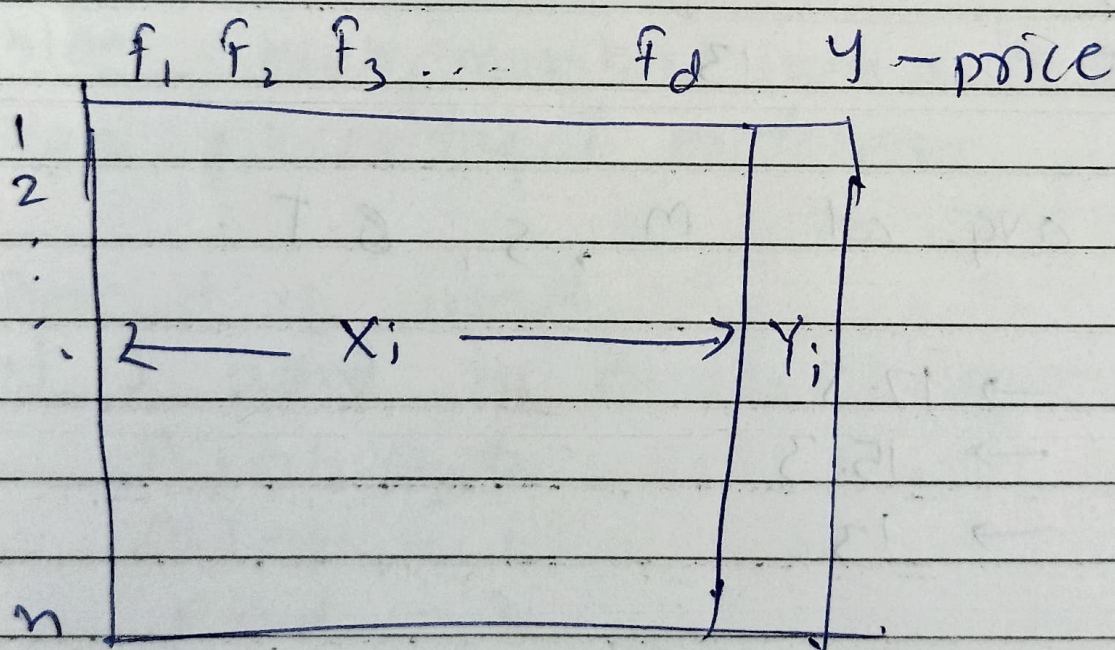
$$S \rightarrow 15.3$$

$$T \rightarrow 13$$

ⓑ Above is ~~kind of~~ similar to
table encoding high number
will give high preference.

It's ok, as it is not random
~~label encoding~~ label encoding does it
randomly which is not good in
this case

Data Notation



m = no. of samples

d = no. of feature

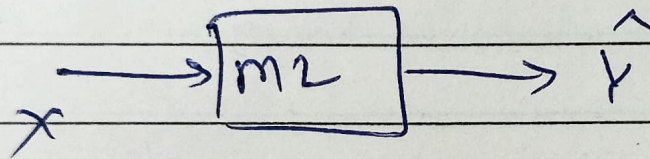
n = no. of classes (o/p classes)

$X(m, d)$

$$x' = [x'_1, x'_2, \dots, x'_d]$$

$x \rightarrow \text{true}$

$\hat{y} \rightarrow \text{predicted}$



$$\hat{y} \approx y$$

to check if model is good we have to test with unknown data (not used in training)

